Material coordinates



Bulk energy W(F, z)

Properties move with the object

Tool-relative attachments



Attachment energy + slip state

Known tool motion; fitted coupling

World-space interactions



Gravity + environmental contact

Depend on current world position

Observed prefix → fit scene parameters → freeze → query future tool positions

SoftVTBench

Analytic shape + position correction
Slip / release + reduced contact rollout

PhysTwin

Frozen neural bulk + scene descriptors
Equilibrium + shared Gaussian appearance

Common physical roles; benchmark-specific adapters. Diagrams are schematic.